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SCREEN ADJUSTER FOR A MOTOR VEHICLE

Abstract

A screen adjustment device for a motor vehicle is provided that comprises a screen reversibly movable between a parking position and at least one operating position and/or screen support for mounting such a screen. At least one electromotive drive unit generates an actuating force which causes an adjustment of the screen and/or screen support. A positioning mechanism guides the screen and/or screen support between the parking position and at least one operating position and transmits the actuating force. The positioning mechanism having at least one spindle drive with a threaded spindle and with a spindle nut arranged on it. The spindle drive being coupled to the screen and/or to the screen support and to the drive unit.

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Background/Summary

[0001] This nonprovisional application claims priority under 35 U.S.C. § 119 (a) to German Patent Application No. 10 2024 201 965.8, which was filed in Germany on Mar. 1, 2024, and which is herein incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to a screen adjustment device for a motor vehicle, in particular for a vehicle interior.

Description of the Background Art

[0003] In modern motor vehicles, a screen (dash display) is often provided in the area of the dashboard or the center console, which serves as the output interface of an on-board computer to display a wide range of environmental and operating variables, such as the outside temperature, fuel consumption, etc. In addition or alternatively, such a screen often serves as an output device for a navigation system, car radio or similar. If a touch-sensitive screen is used or if separate controls are assigned to a display that is not touch-sensitive in itself, such a screen can also serve as an input interface for the on-board computer, the navigation system, or the car radio.

[0004] Such screens are preferably installed in the dashboard or in the center console so as to swivel or tilt, so that the screen can be swiveled or tilted sideways towards both the driver and the front passenger. This allows for increased flexibility and improved visibility for both occupants, which significantly increases operating comfort and safety while driving as compared to fixed screens.

[0005] Swiveling screens are often adjustable by means of a screen adjustment device (screen adjuster) in such a way that they are swiveled from a resting or parking position in which they are adjacent to the dashboard. With such a swivel movement, it is necessary for the screen to be moved at least partially out of the dashboard, i.e., in the direction of the vehicle interior, otherwise part or a section of the screen is swiveled into the interior of the dashboard, thus adversely affecting the full visibility and accessibility of the screen. It is therefore necessary to move motor vehicle screens along an, often complex, adjustment travel between the parking position and an operating position-which is swiveled in the vehicle interior. To move to the operating position, the screen is usually pushed out of the dashboard in an essentially horizontal direction and then, or simultaneously, tilted or swiveled into an operating position inclined sideways towards the driver or front passenger, in which the screen is clearly visible to the vehicle occupant.

[0006] For example, screen adjustment devices are conceivable in which two permanently mounted spindle drives, or spindle adjusters, and an additional lever are used to move and rotate/swivel the screen. The screen is guided in a groove on the back by means of a pin for stabilization.

Disadvantageously, additional installation space and additional components are required to guide the pin, which limits the flexibility of the screen adjustment device due to the limited installation space in the dashboard.

SUMMARY OF THE INVENTION

[0007] It is therefore an object of the invention to provide a particularly suitable screen adjustment device for a motor vehicle. In particular, the aim is to specify a screen adjustment device with as few components and as little installation space as possible.

[0008] The screen adjustment device according to an example of the invention comprises a screen (display, monitor) and/or a screen support (screen holder) on which the screen can be held. The

conjunction “and/or” is to be understood here and in the following in such a way that the features linked by means of this conjunction can be formed both together and as alternatives to each other.

[0009] The screen or screen support can be moved reversibly between a parking position and at least one operating position. In order to generate an actuating force for adjusting the screen or screen support, the screen adjustment device comprises at least one electromotive drive unit.

[0010] Furthermore, the screen adjustment device comprises a positioning mechanism that serves to guide the screen and/or screen support along an adjustment travel between the parking position and the at least one operating position and to transmit the actuating force.

[0011] If the screen support is not provided, the positioning mechanism directly engages the screen, so that the screen is a necessary part of the screen adjustment device. Otherwise, the positioning mechanism engages the screen support, to which the screen is attached. In the latter case, in a convenient design, the screen adjustment device is fully functional even without a screen, so that the screen is not a necessary part of the screen adjustment device. However, it is also conceivable that the positioning mechanism is connected to the screen as well as to a screen support holding or supporting it.

[0012] In any case, the positioning mechanism can have at least one spindle drive. The spindle drive comprises at least one threaded spindle with a spindle nut coupled to it.

[0013] To adjust the screen or screen support, the spindle drive is coupled or connected to the screen and/or screen support on the one hand and to the at least one drive unit on the other. This means that a particularly suitable screen adjustment device has been implemented.

[0014] The positioning mechanism can have a rail arrangement oriented along the linear adjustment range, in particular, with a bottom rail (guide rail) for coupling to the screen and/or screen holder and an top rail (guide slide) for coupling to the drive unit that is seated in the bottom rail in a form-fitting and relatively movable manner. In other words, a linear drive is provided to adjust the screen and/or the screen holder.

[0015] The threaded spindle is fixed to the bottom rail, with the spindle nut being connected to the top rail in terms of drive technology. For example, the drive unit may be directly or indirectly connected to the top rail. For example, a joining or mounting plate that can be connected to the top rail is provided, to which the drive unit and the screen and/or the screen holder are attached.

[0016] Preferably, a rail arrangement is used as described in the applicant's German application 10 2022 207 168.9, which corresponds to PCT/EP2022/082230, which is incorporated herein by reference. Its disclosed content, in particular its claims (with related explanations), are hereby expressly included in the present application, wherein the statements regarding the longitudinal seat adjustment are transferable mutatis mutandis to the screen adjustment. The content of this application is incorporated by reference as an integral part of the present application. Reference is made specifically to the feature that the top rail has additional track rollers arranged at an angle, which are spring-loaded on obliquely oriented side walls of the bottom rail.

[0017] The top rail can be arranged in a fixed position, wherein the bottom rail is displaced relative to this by the spindle drive. The bottom rail is connected to the screen and/or the screen support, for example, by mounting flanges, wherein the top rail is fixed to a frame of the dashboard or center console, for example, via a mounting plate.

[0018] This ensures play-free guidance along the adjustment travel formed by the bottom rail. Due to the freedom from play, fluttering or rattling noises are reliably avoided when adjusting the screen.

[0019] The linear drive, for example, is designed according to the modular principle, so that many identical parts can be used for the different applications (screen adjustment, seat adjustment, . . .). The variance arises only due to the “bottom rail” length and specific connection parts.

[0020] The screen adjustment device according to the invention can be designed with a laterally swiveling screen and/or screen holder.

[0021] The drive unit can be coupled to the spindle nut and the threaded spindle to the screen

and/or to the screen support. In other words, the spindle nut is driven by the drive unit in order to adjust the screen and/or screen support, with the threaded spindle coupled to the screen and/or screen support and retracted and extended relative to the spindle nut.

[0022] The threaded spindle can be directly connected to the screen and/or to the screen support at the end in an articulated or movable manner. The threaded spindle can be connected to the screen and/or the screen support by means of a swivel joint, so that the screen and/or the screen support can be swiveled or tilted to the side.

[0023] In a practical design, the positioning mechanism can have a longitudinal adjustment unit with a bracket and with at least one motor cage fixed to it, wherein the at least one drive unit is rotatable in the motor cage. The motor cage is the stationary bearing for the spindle nut driven by the drive unit. The drive unit being rotatable on the motor cage makes it possible for the threaded spindle to be swiveled relative to the bracket or motor cage without affecting the drive or the actuating force.

[0024] The positioning mechanism or the longitudinal adjustment unit can have two hinged threaded spindles coupled to the screen and/or the screen support and two motor cages fixed to the bracket. In each of the motor cages, a rotatable drive unit is mounted, which is coupled with a spindle nut for adjusting one of the threaded spindles in terms of drive technology. The spindle nuts are also located in the motor cages.

[0025] In this design, the screen and/or the screen support can be mounted directly on two threaded spindles, which are driven by two swiveling spindle adjusters. The more direct drive of the screen and/or screen support by the spindle drives reduces the number of components (no additional coupling). As a result, the production costs and the installation space required for installation are reduced. To be more precise, a larger adjustment angle for the screen and/or the screen support can be realized in the same installation space. The reduced number of components results in less susceptibility to play and sensitivity to vehicle vibrations.

[0026] The bracket can be directly or indirectly hinged to the screen and/or to the screen support. Preferably, the bracket is pivotally connected to the screen and/or the screen support via a swivel joint. This ensures stable and reliable guidance of the screen and/or screen support.

[0027] For example, the bracket of the longitudinal adjustment unit between the spindle adjusters can be designed as a rail guide. For example, the rail guide can be designed in the manner of the applicant's German application of Jul. 13, 2022. The rail guide has a stationary or fixed bottom rail, and a top rail guided on it, wherein the top rail is pivotally connected to the screen and/or the screen support at the end by a swivel joint. The stability of the screen and/or the screen support, especially in the extended position, can be increased with the hinged top rail.

[0028] A guide groove can be inserted into the bracket, in which a guide element connected to the screen and/or to the screen support is guided in a movable manner. The guide groove as a guide track stabilizes the screen and/or the screen support when these are extended. The risk that the guide element takes a wrong path in the guide groove is virtually eliminated.

[0029] In an expedient design, the guide groove can have an essentially V-shaped course in the bracket. The two V-legs correspond to the two travel or adjustment ranges for swiveling in the direction of the driver or passenger, wherein the V-base (i.e., the point of contact of the two V-legs) corresponds to the retracted resting position.

[0030] The screen and/or the screen support can be moved between the parking position and two operating positions by means of the positioning mechanism, wherein the two threaded rods are moved different distances in the operating positions.

[0031] Depending on the ratio of the travel or adjustment range of the threaded spindles, the screen and/or screen support will move away from the bracket and swivel left or right at the same time. When installed in a dashboard, the screen and/or the screen support thus moves out of the dashboard and at the same time swivels towards the driver or passenger. In other words, the screen and/or the screen support with the spindles move at least partially out of the dashboard, so that the

visibility of the screen is not impaired when swiveling. An additional guide (e.g., pin in groove) is not necessary for the movement of the screen and/or screen support but may be provided for additional stabilization. For swiveling, for example, one of the threaded spindles is extended only half as far as the other threaded spindle.

[0032] In the event of a power failure, it is desirable that an extended or swiveled screen and/or screen support can be manually moved back to the resting position (parking position) or into the dashboard. For this purpose, it is provided in a suitable further development that the at least one drive unit is designed by means of gearing and transmission in such a way that it can be driven backwards in the direction of the parking position. In emergency operation, manual retraction is thus implemented via a non-self-locking drive unit.

[0033] Further scope of applicability of the present invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes, combinations, and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus, are not limitive of the present invention, and wherein:

[0035] FIG. 1 shows a screen adjustment device in an operating position in a perspective representation,

[0036] FIG. 2 shows the screen adjustment device in a parking position in a perspective representation,

[0037] FIG. 3 shows an adjustment unit of the screen adjustment device with mounting flanges in a perspective representation,

[0038] FIG. 4 shows the adjustment unit without the mounting flanges in a perspective representation,

[0039] FIG. 5 shows the adjustment unit in a disassembled state in a perspective exploded representation,

[0040] FIG. 6 shows the adjustment unit in a side view,

[0041] FIG. 7 shows a screen adjustment device in accordance with an example in a parking position in a plan view,

[0042] FIG. 8 shows the screen adjustment device in accordance with the example in the parking position in a perspective representation,

[0043] FIG. 9 shows the screen adjustment device in accordance with the example in an operating position in a plan view,

[0044] FIG. 10 shows the screen adjustment device in accordance with the example in the operating position in a perspective representation,

[0045] FIG. 11 shows a screen adjustment device in accordance with an example in a parking position in a plan view, and

[0046] FIG. 12 shows the screen adjustment device in accordance with the example in an operating position in a plan view.

DETAILED DESCRIPTION

[0047] The screen adjustment device 2 shown in FIG. 1 is intended in particular for installation in a dashboard or in a center console of a motor vehicle. In particular, the screen adjustment device 2 is

arranged approximately in the middle between a driver's seat and a front passenger seat.

[0048] In the following, information regarding spatial directions is also given, in particular in a coordinate system of the motor vehicle (vehicle coordinate system) with regard to an exemplary installation situation of the screen adjustment device **2** in a dashboard. The abscissa axis (X-axis, X-direction) is oriented along the longitudinal direction of the vehicle (direction of travel) and the ordinate axis (Y-axis, Y-direction) along the transverse direction of the vehicle, and the applicate axis (Z-axis, Z-direction) is oriented along the height of the vehicle.

[0049] The screen adjustment device **2**, for example, has a screen **4** permanently connected to it as a vehicle display. Instead of a screen **4** permanently connected to the screen adjustment device **2**, the screen adjustment device **2** may also comprise a screen support **6** to which a screen **4** can be attached retroactively. In the latter case, the screen adjustment device **2** is fully functional even without a screen **4**. It is also conceivable that the screen adjustment device **2** is connected both to a screen support **6** and to a screen **4** held by it.

[0050] In the example of FIGS. **1** to **6**, the screen adjustment device **2** has a screen support **6** and a screen **4** mounted on it. The screen support **6**—and thus the screen **4**—can be moved reversibly relative to the dashboard by means of an adjustment unit **8**. When the adjustment unit **8** is activated, the screen support **6** or the screen **4** is moved between a parking position P (FIG. **2**) recessed in the dashboard and an extended operating position B (FIG. **1**).

[0051] The adjustment unit **8**, which is explained in more detail below on the basis of FIGS. **3** to **6**, has an electromotive drive unit **10** and a positioning mechanism **12**.

[0052] The positioning mechanism **12** has a spindle drive **14** and a rail arrangement **15**.

[0053] The spindle drive **14** comprises a threaded spindle **16** and a spindle nut **18** moveably guided thereon. The spindle nut **18** is in a drive coupling with a drive shaft **20** of the drive unit **10**. When the drive unit **10** is actuated, the spindle nut **18** is driven, and is thus adjusted along the threaded spindle **16**.

[0054] The rail arrangement **16** has a top rail **22** and a bottom rail **24** guided on it so that it can be moved.

[0055] The top rail **22** is designed as a guide carriage with a roughly U-shaped carriage body **26**. The horizontal U-leg has two mounting openings **28** for coupling with the spindle nut **18** and with the drive unit **10**.

[0056] The carriage body **26** has an essentially three-part structure along the longitudinal direction of the body. In the middle area of the carriage body **26**, four load-bearing load rollers **30** are arranged, which are arranged in pairs and rotatable on one of the vertical U-legs opposite each other. In the outer sections of the carriage body **26**, the vertical U-legs are bent outwards and upwards as spring tabs **32** in the direction of the horizontal U-leg. A rotatable track roller **34** is attached to each of the spring tabs **32**, arranged on the free-end side. The spring tabs **32** are preferably flexible or resilient.

[0057] In the horizontal U-leg, between the mounting openings **28**, a window-like recess **36** is incorporated for the spindle nut **18**, wherein the recess **36** is coupled with a slot **38** extending in sections over the vertical U-leg on the drive side.

[0058] The spindle nut **18** is pivotably or rotatably mounted in a cage-like bracket **40**. The bracket **40** is essentially U-shaped, with the free ends of the vertical U-legs bent outwards at right angles as mounting tabs **42**. The mounting tabs **42** each have a mounting opening **44**.

[0059] Above the spindle nut **18**, a drive element **46** is arranged in the bracket **40**. The drive element **46** has, for example, a worm shaft meshing with the external thread of the threaded nut **18**, which is connected to the drive shaft **20** in the assembled state.

[0060] For assembly, the bracket **40** is inserted from below into the carriage body **26** of the top rail **22**, so that the drive element **44** sits in the recess **36**, and so that the mounting openings **44** are arranged in alignment with the mounting openings **28**. Subsequently, the drive shaft **20** is coupled laterally through the slot **38** to the drive element **46**.

[0061] The top rail **22** can be permanently connected to the drive unit **10** via a mounting plate **48**. For this purpose, a drive housing **50** of the drive unit **10** has side-molded mounting flanges **52**, which can be fixed to the mounting plate **48** by fasteners **54**.

[0062] The mounting plate **48** also has a window-like recess **56** and two mounting openings **58**. Recess **56** is aligned with recess **36**, with mounting openings **58** aligned with mounting openings **28** and **44**. The top rail **22**, the bracket **40** and the mounting plate **48** are firmly connected by two fasteners **60** that penetrate the mounting openings **28**, **44**, **58**.

[0063] The approximately U-shaped bottom rail **24** has a rail floor **62** as a horizontal U-leg and two side walls **64** as vertical U-legs. The free ends of the side walls **64** are angled diagonally inwards. The threaded spindle **16** is non-rotatably attached to the rail floor **62** of the bottom rail **24** by means of two fastening brackets **66**.

[0064] The bottom rail **24** is coupled to the screen support **6** via two mounting end plates **68**. The mounting end plates **68** are arranged vertically to the outside on the end of the rail on the side walls **64**.

[0065] The top rail **22** and the drive unit **10** are mounted stationary or fixed when installed. For this purpose, a mounting plate **70** with four mounting tabs **72** is provided, which is firmly connected to the top rail **22**, the bracket **40** and the mounting plate **48** via fasteners **60** (FIG. 3, FIG. 6).

[0066] When assembled, the top rail **22** and bottom rail **24** are oriented along the vehicle height direction Z. As can be seen in particular in FIG. 6, the top rail **22** is seated in a form-fitting manner in the bottom rail **24**. The positive fit is oriented along a vertical rail height direction (i.e., along the vehicle's longitudinal direction X) and along a rail transverse direction (i.e., along the vehicle's transverse direction Y) of the rail arrangement **15**. The load rollers **30** of the top rail **22** sit on the rail floor **62**, wherein the diagonally oriented track rollers **34** of the spring tabs **32** are supported as running surfaces on the diagonally inwardly directed sections of the side walls **64**.

[0067] By supporting the top rail **22** at the same time by means of the load rollers **30** and the track rollers **34** on the bottom rail **24**, a play-free guidance along the adjustment range formed by the bottom rail **24** is realized. Due to the freedom from play, fluttering or rattling noises when adjusting the screen **4** are reliably avoided.

[0068] A second example of the screen adjustment device **2** is explained in more detail below using FIGS. 7 to 10. In this example, the screen **4** and/or the screen support **6** can be swiveled sideways by means of the screen adjustment device **2**.

[0069] The screen adjustment device **2** is adjustable between a parking position P and two sideways swiveling operating positions B1, B2, wherein one operating position is swiveled to the driver's seat and the other operating position to the front passenger seat. In the figures, only one operating position is depicted, the other is essentially the result of a reflection. In the parking position P, the screen **4** is arranged parallel and preferably visually aligned with a visible area of the dashboard.

[0070] In this example, the adjustment unit **8** has two drive units **10**, wherein the positioning mechanism **12** is designed with two spindle drives **14** and with a longitudinal adjustment unit **74** arranged in between.

[0071] The longitudinal adjustment unit **74** has a bracket **76**. The bracket **76** comprises a stationary or fixed mounting plate **78** on which two fixed motor cages **80** are arranged. The motor cages **80** are designed for the support of the two drive units **10** as well as for the stationary and rotatable bearing of the spindle nuts **18**.

[0072] As can be seen in particular in the illustration of FIG. 10, the longitudinal adjustment unit **74** or the bracket **76** has a rail arrangement **15**, wherein a top rail **22** is fixed on the mounting plate **78**, and a bottom rail **24**, which is arranged on it in a movable manner, is hinged to the screen support **6** by means of a swivel joint **82**. The linear rail guidance stabilizes the extension of the screen **4** or screen support **6** into the vehicle interior, with the swivel joint **82** forming the swivel axis for the swivel movement towards the driver's seat or front passenger seat.

[0073] The drive units **10** are mounted on the respective motor cages **80** in a rotatable manner. The

drive units **10** can thus rotate along a rotary or swivel axis oriented essentially parallel to the drive shaft **20** relative to the fixed motor cage **80**. For this purpose, the motor cage **80** has a recess in the shape of a kidney or bean as a guide contour **84** for the drive shaft **20**.

[0074] The threaded spindles **16** are each coupled to the screen support **6** via a spindle-end-side swivel joint **86**. The threaded spindles **16** can be swiveled via the swivel joints **86** relative to the screen support **6**. However, along the spindle longitudinal direction, the threaded spindles **16** are arranged in a non-rotatable manner.

[0075] When adjusting the screen adjustment device **2** from the parking position P to one of the operating positions B1, B2, the drive units **10** are activated, thus driving the spindle nuts **18**. This moves the threaded spindles **16** along the X-direction, thereby moving the bottom rail **24** along the top rail **22** so that the screen **4** or the screen support **6** is moved out from the dashboard.

[0076] In a conceivable design, one of the drive units **10** is then stopped, while the other drive unit **10** continues to drive the assigned spindle nut **18**. This continues to move this side further along the X direction. In an example, however, both drive units **10** are operated at different adjustment speeds, and essentially arrive at their end positions at the same time.

[0077] By means of the swivel joints **86** and **82**, the screen support **6** and the screen **4**, respectively, are swiveled laterally around the swivel axis formed by the swivel joint **82**. As a result, the swivel joint **86** of the non-driven or slower spindle drive **14** is swiveled inwards, wherein the articulated bearing tilts or swivels the threaded spindle **16** parallel to the Z-direction (cf. e.g., FIG. 9). The swivel of the threaded spindle **16** is limited by the rotation of the stationary or slower drive unit **10**, which is limited in the guide contour **84**. The drive shaft **20**—and thus the drive unit **10**—is swiveled along the guide contour **84** when the threaded spindle **16** is moved due to the rotatable drive unit. As a result, the threaded spindle **16** of the non-active or slower spindle drive **14** is swiveled relative to the bracket **76** or the motor cage **80** without affecting the drive unit **10** or the actuating force.

[0078] When extending, the drive shafts **20** or the gearbox housing of the drive unit **10** thus rotate in the motor cage **80**, so that a changing distance of the spindle eye forming the swivel joint **86** to the vehicle center due to the swiveling movement is compensated. The swivel movement indicated in FIG. 9 with a double arrow **87** is not limited, and results from the adjustment movement of the screen **4**.

[0079] In a suitable dimensioning, it is necessary, for example, for the adjustment or movement from the parking position P to the operating position B1, that one drive unit **10** has an adjustment range of 100 mm (millimeters) and the other drive unit **10** has an adjustment range of 50 mm. With a target travel time of 4 s (seconds), the first drive unit travels at an adjustment speed of 25 mm/s (millimeters per second), and the other drive unit at 12.5 mm/s. This makes it look as if the screen **4** only tilts out of the dashboard by the edge of the screen in the vehicle interior (cf. e.g., illustrations of FIG. 7 and FIG. 9).

[0080] The advantage of the swivel adjustment of the example in FIG. 7 to FIG. 10 is that a desired adjustment angle can be swiveled to a fixed adjustment angle (e.g.,) 20°. Thus, for example, any desired adjustment angle (e.g., 10°, 15°, . . .) can be set by appropriate software and/or control of the electromotive drive units **10**. Furthermore, it is possible to easily adapt to different screen dimensions (display sizes) without having to change parts of the screen adjustment device **2** geometrically.

[0081] A further advantage is that the screen adjustment device **2** in FIG. 7 to FIG. 10 allows for direct swiveling between the operating positions B1, B2 by one drive unit **10** running forward and the other drive unit **10** backwards. This eliminates the need to temporarily return to the parking position P.

[0082] A third example of the screen adjustment device **2** is explained in more detail below on the basis of FIG. 11 and FIG. 12. This example is essentially the same as the example in FIGS. 7 to 10 described above, in which the rail arrangement **15** of the longitudinal adjustment unit **74** is replaced

by a groove guide **88**.

[0083] The groove guide **88** has a guide groove **90** as a guide track for a guide element **92** designed as a (guide) pin. The guide groove **90** is designed as a V-shaped recess of the mounting plate **78** in the XY-plane, wherein the V-legs are slightly curved or bent outwards. The guide element **92** sits in the guide groove **90** in a form-fitting and movable manner and is firmly connected to the back of the screen **4** or the screen support **6** via a bracket **94**.

[0084] The guide element **92**, which can be rotated in the guide groove **90**, also realizes a swivel joint **96** for the lateral swiveling of the screen **4** and/or the screen support to the operating position **B1**, **B2** (FIG. **12**).

[0085] In FIG. **11**, the screen adjustment device **2** is located in the parking position **P**. The guide element **92** is located at the V-base of the V-shaped guide groove **90**, which corresponds to the farthest point of the guide groove **90** to the screen **4** and/or screen support **6**.

[0086] In order to move the screen adjustment device **2** from the parking position **P** to the operating position **B1**, **B2** (or back), both drive units **10** are preferably operated at different adjustment speeds.

[0087] Alternatively, the drive unit **10** is activated first, which is located on the side on which the spindle drive **14** is further extended. This ensures that the guide element **92** slides into the V-leg of the guide groove **90** assigned to the desired operating position **B1**, **B2**. In particular, the guide element **92** is pulled into the V-leg by the bracket **96** when the drive element **10** drives the spindle drive **14** and adjusts the threaded spindle **16**.

[0088] The second drive unit **10** is then also activated until the screen support **6** or screen **4** is extended sufficiently far out of the dashboard. The first drive unit **10** adjusts the corresponding spindle drive **14** until the guide element **92** has reached the free end of the V-leg of the guide groove **90**.

[0089] During the adjustment, the screen **4** and/or screen support **6** tilts or swivels by rotating the guide element **92** around the axis of rotation formed by the guide element **92** or the swivel joint **96**. The curvature or arch of the V-leg of the guide groove **90** is dimensioned accordingly so that no unnecessary tension or stress acts on the guide element **92**.

[0090] The example of the guide groove **90** as a V-contour with a curved guide track is only one conceivable possibility. The type of guide track or guide groove **90** depends on how the screen **4** should rotate when it is moved out of the dashboard. It is therefore also possible to make the guide track straight (linear). In terms of its geometry, the guide groove **90** is preferably adapted to the desired adjustment kinematics and, above all, to the screen size.

[0091] The claimed invention is not limited to the examples described above. Rather, other variants of the invention can also be derived by the skilled person within the framework of the disclosed claims without departing from the subject-matter of the claimed invention. In particular, all the individual features described in connection with the various examples can also be combined in other ways within the framework of the disclosed claims without departing from the subject-matter of the claimed invention.

[0092] The motor speeds of the drive units **10** are controlled accordingly to ensure that the screen swivels in and out smoothly and comfortably. Preferably, the drive units **10** have BLDC electric motors.

[0093] The screen adjustment devices **2** in FIGS. **1** to **12** are preferably designed in such a way that, in the event of a power failure, an extended or swiveled screen **4** and/or screen support **6** can be manually moved back to the resting position (parking position **P**) or into the dashboard. For this purpose, it is provided that each drive unit **10** and each spindle drive **14** are designed with a gearing and transmission in such a way that they can be driven backwards in the direction of the parking position **P**. In emergency operation, manual retraction is thus implemented via a non-self-locking drive unit **10** or a non-self-locking positioning mechanism **12**.

[0094] In the case of the screen adjustment devices **2** of the examples described above, in each case

one screen **4** and one screen support **6** are provided. However, the screen adjustment devices **2** may also be designed with only a screen **4** or with only a screen support **6**.
[0095] The invention being thus described, it will be obvious that the same may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the invention, and all such modifications as would be obvious to one skilled in the art are to be included within the scope of the following claims.

Claims

- 1.** A screen adjustment device for a motor vehicle, the screen adjustment device comprising: a screen and/or a screen support for supporting the screen adapted to be reversibly moved between a parking position and at least one operating position; at least one electromotive drive unit to generate an actuating force which causes the screen and/or screen support to be adjusted; and a positioning mechanism for guiding the screen and/or screen support between the parking position and the at least one operating position as well as for transmitting the actuating force, wherein the positioning mechanism has at least one spindle drive with a threaded spindle and with a spindle nut arranged on it, and wherein the spindle drive is coupled with the screen and/or with the screen support and with the drive unit (**10**).
 - 2.** The screen adjustment device according to claim 1, wherein the drive unit is coupled to the spindle nut and the threaded spindle to the screen and/or to the screen support.
 - 3.** The screen adjustment device according to claim 2, wherein the threaded spindle is directly connected to the screen and/or to the screen support at an end.
 - 4.** The screen adjustment device according to claim 2, wherein the positioning mechanism has a longitudinal adjustment unit with a bracket and with at least one motor cage fixed thereto, and wherein the at least one drive unit is rotatably mounted in the motor cage.
 - 5.** The screen adjustment device according to claim 4, wherein the positioning mechanism has two threaded spindles articulated to the screen and/or the screen support and two motor cages arranged permanently on the bracket, wherein, in each of the motor cages, a drive unit is rotatably mounted, which is coupled in terms of drive technology with a spindle nut for adjusting each of the threaded spindles.
 - 6.** The screen adjustment device according to claim 4, wherein the bracket is articulated to the screen and/or to the screen support.
 - 7.** The screen adjustment device according to claim 6, wherein a guide groove is incorporated in the bracket in which a guide element connected to the screen and/or to the screen support is guided in a movable manner.
 - 8.** The screen adjustment device according to claim 7, wherein the guide groove has an essentially V-shaped course in the bracket.
 - 9.** The screen adjustment device according to claim 5, wherein the screen and/or the screen support are adapted to be moved by the positioning mechanism between the parking position and two operating positions, and wherein the two threaded rods are moved different distances in the operating positions.
 - 10.** The screen adjustment device according to claim 2, wherein at least one drive unit is designed such that it is adapted to be driven backwards in a direction of the parking position via a gearing and transmission.
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